



SIZE SEPARATION

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Size separation

- ❖ In pharmaceutical industry, the control of particle size and size range is of great importance
- ❖ Various methods of size reduction is used but it is not necessarily that they will produce the desired size particles and of uniform size

Objectives of size separation

- ❖ *To facilitate the flowability of powders*
- ❖ *To facilitate the proper mixing of granules and powders*
- ❖ *To ensure uniformity of dose*
 - To do this, the powder particles should have particle size distribution with a narrow range

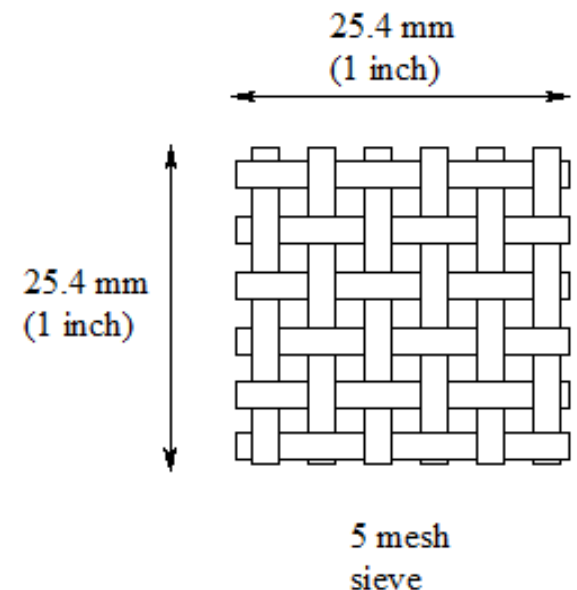


Size separation...

- ❖ The successful formulation of DFs from the view points of both physical stability and pharmacological response depends on the particle size achieved in the product

Size separation by Sieving

- ❖ Sieving/sifting is a process by which d/t grades of powders are separated from each other
- ❖ Sieves are made up of wood or metal which is fitted with a wire cloth made of copper, steel, silk or nylon
- ❖ *Each sieve is given a number which indicates the number of meshes of standard diameter present in length of 1 inch,*
 - **E.g. no 5 sieve has 5 meshes in 1 inch in each direction**

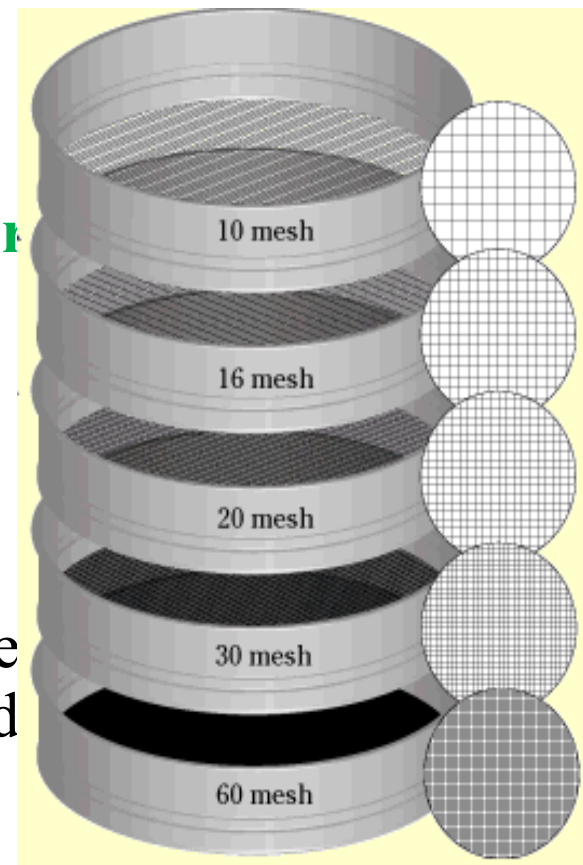


Size separation by Sieving...

- ❖ Sieving is the most widely used method for size separation and measuring the particle size distribution because it is:
 - **Inexpensive**
 - **Simple and rapid**
 - **Little variation b/n operators**
- ❖ Sieving involves the mechanical shaking of a sample through a series of successively smaller sieves and the weighing of the portion of the sample retained on each sieve

Size separation by Sieving...

- ❖ Sieve of d/t sizes are arranged simultaneously one above the other and fit into each other in such a way that **the coarsest sieve is on the top followed by other sieves in increasing order of their mesh number**
- ❖ *The particles smaller than the mesh size will pass through the sieve and over size particles will be retained on the sieve.*
- ❖ After shaking for required period of time the weight of each fraction is determined and the percentage is calculated.



Size separation by Sieving...

- ❖ Size analysis by sieving is influenced by:
 - **The type of motion**
 - **Time of sieving**
 - **Load or thickness of powder**
- ❖ Sieves should be used and stored with care, since a sieve is of little value if the meshes become damaged
- ❖ Particles, if small enough, will pass through a sieve easily if it is shaken, tapped or brushed
- ❖ There are two methods of sieving
 - Mechanical sieving
 - Wet sieving

Size separation by Sieving...

Mechanical sieving method

- ❖ Mechanical sieving devices are usually based on methods that agitate or brush the sieve or use centrifugal force

1. Agitation methods:-

- ❖ Sieves may be agitated in a number of d/t ways:
 - **Oscillation**
 - **Vibration**
 - **Gyration**

2. Brushing methods

- ❖ In large scale production, a horizontal cylindrical sieve is employed with a spiral brush rotating on the longitudinal axis of the sieve

Size separation by Sieving...

3. Centrifugal methods

- ❖ Mechanical sieves of this type normally use a vertical cylindrical sieve with a high speed rotor inside the cylinder so that particles are thrown outwards by centrifugal force.

Size separation by Sieving...

Standards or grading of powders

- ❖ Standards of powders for pharmaceutical purposes are laid down principally in the **BP** which states that the degree of coarseness or fineness of powder is differentiated and expressed by the size of the mesh of the sieve through which the powder is able to pass
- ❖ The **BP** specifies five grades of powder which gives the name by which the grade of powder is known and the number of the sieve through which all the particles must pass

Size separation by Sieving...

<i>BP powder</i>	<i>sieve through w/c</i>	<i>sieve through w/c</i>
<i>Grades</i>	<i>all particles pass</i>	<i>NMT 40% of particles pass</i>
<i>Coarse</i>	<i>10</i>	<i>44</i>
<i>Moderately coarse</i>	<i>22</i>	<i>60</i>
<i>Moderately Fine</i>	<i>44</i>	<i>85</i>
<i>Fine</i>	<i>85</i>	<i>not specified</i>
<i>Very Fine</i>	<i>120</i>	<i>not specified</i>

❖ What is coarse powder?